Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Regula Falsi Method lies in the category of ----------

Answer (Please select your correct option)

- Iterative method
- Bracketing method
- Random method
- Graphical method

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Secant method converges --------than bisection

convergence is faster than bisection.

Answer (Please select your correct option)

- Faster
- Slower
- Equally
- None of the given choices

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Muller's method is a generalization of

Muller's method is a generalization of the secant method, in the sense that it does not require the derivative of the function. It is an iterative method that requires three starting points $(p_0, f(p_0))$, $(p_1, f(p_1))$, and $(p_2, f(p_2))$. A parabola is constructed that passes through the three points; then the quadratic formula is used to find a root of the quadratic for the next approximation.

Answer (Please select your correct option)

- Bisection method
- Iteration method
- Secant method ✓ ▭ ⇦
- Regula Falsi method

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

Diagonal dominance of a coefficient matrix is strictly checked in

we assume that the coefficient matrix [A] is strictly diagonally dominant

Answer (Please select your correct option)

- Muller's method.
- Bisection method
- Jacobi's method ✓ ⇦ pg 69 71
- Newton-Raphson method

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

It can be verified that for matrix $A=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix}$

Answer (Please select your correct option)

- $AA^{-1} = I$, I = identity matrix
- $AA^{-1} = D$, D = diagonal matrix
- $AA^{-1} = S$, S = symmetric matrix ✓ ⇐
- $AA^{-1} = Z$, Z = orthogonal matrix

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If $[A]$ is an $n \times n$ real symmetric matrix, its eigen values are real, and there exists an orthogonal matrix $[S]$ such that the diagonal matrix $[D]$ is given by

if $[A]$ is an $n \times n$ real symmetric matrix, its eigen values are real, and there exists an orthogonal matrix $[S]$ such that the diagonal matrix D is
$[S^{-1}][A][S]$

Answer (Please select your correct option)

- $[D] = [S]^{-1}[A][S]$ ✓ pg104
- $[D] = [S]^{-1}[S][A]$
- $[D] = [S]^{2}[A][S]$
- None of the above

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

The symbol used for average operator is

Answer (Please select your correct option)

- Δ
- μ ✓ ⇦ pg 265 267
- ∇
- δ

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	4	5	7	10
$f(x)$	48	100	294	900

uncomplete question

Answer (Please select your correct option)

- 94
- 97
- 194
- 52

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 1/3 rule, the number of intervals in the following must be

1/3rule intervals 6 and 3/8rule intervals 9

Answer (Please select your correct option)

6 R ✓

7

9

11

The Simpson's 1/3 rule, we have used two sub-intervals of equal width. In order to get a composite formula, we shall divide the interval of integration [a, b] Into an even number of sub- intervals say 2N, each of width (b – a)/2N, thereby we have

$x_0 = a, x_1, \ldots, x_{2N} = b$ and $x_k = x_0 + kh$, $k = 1,2, \ldots (2N - 1)$.

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_1^2 x^2 dx$, by dividing the interval into eight equal parts, width of the interval should be

Repeat

Answer (Please select your correct option)

- 0.125 R ✓
- 0.250
- 0.500
- 0.625

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Marks: 1 (Budgeted Time 1 Min)

Question No : 15 of 52

P in Newton's backward difference formula is defined as

pg 222

Answer (Please select your correct option)

- $p = (\frac{x - x_0}{h})$
- $p = (\frac{x + x_0}{h})$
- $p = (\frac{x + x_n}{h})$
- $p = (\frac{x - x_n}{h})$ R ✓

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Marks: 1 (Budgeted Time 1 Min)

Question No : 16 of 52

Given the following data

x	1	2	-4
$f(x)$	3	-5	-4

Which formula is useful in finding the interpolating polynomial?

Answer (Please select your correct option)

- Newton's backward difference formula
- Lagrange's interpolation formula
- None of the given choices
- Newton's forward difference formula

intervals are not equally spaced

R ✓

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

using two-point equation the value of f '(0.2) is.....

Answer (Please select your correct option)

- 0.75
- -0.75 R ✓
- 0.5
- -0.5

$$f'(0.1) = \frac{f(0.2) - f(0.1)}{h} = \frac{0.475 - 0.425}{0.1} = \frac{0.050}{0.1} = 0.5.$$

$$f'(0.2) = \frac{f(0.3) - f(0.2)}{h} = \frac{0.400 - 0.475}{0.1} = \frac{-0.075}{0.1} = -0.75.$$

$$f'(0.3) = \frac{f(0.4) - f(0.3)}{h} = \frac{0.450 - 0.400}{0.1} = \frac{0.05}{0.1} = 0.5.$$

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If $y(x)$ is approximated by a polynomial $p_n(x)$ of degree n then the error is given by

Answer (Please select your correct option)

- $\varepsilon(x) = y(x) + P_n(x)$
- $\varepsilon(x) = y(x) - P_n(x)$ R ✓
- $\varepsilon(x) = P_n(x) - y(x)$
- $\varepsilon(x) = y(x) \times P_n(x)$

Error Term in Interpolation Formulae

We know, if y (x) is approximated by a polynomial P_n (x) of degree n then the error is given by

$$\varepsilon(x) = y(x) - P_n(x),$$

Where

$$\varepsilon(x) = (x - x_0)(x - x_1)\cdots(x - x_n)y[x, x_0, \ldots, x_n]$$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

If $f(x) = 5x^6 + 6x^5 - 7x^3 - 9x^2 + 4x - 3$, then its------derivative is zero for all x.

is a polynomial of degree n, its (n + 1)th derivative is zero

Answer (Please select your correct option)

- 4th
- 7th R ✓
- 6th
- 5th

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

From the following table of values:

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y	1.0000	1.0247	1.0488	1.0724	1.0954	1.1180	1.1402

question incomplete

Answer (Please select your correct option)

- Forward difference operator
- Backward difference operator
- Central difference operator
- None of the given choices

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

In Richardson's extrapolation method, the extrapolation process is repeated until accuracy is achieved, this is called extrapolation to the .....

Answer (Please select your correct option)

- limit

R ✓ This extrapolation process can be repeated further until the required accuracy is achieved, which is called an extrapolation to the limit. Therefore the equation for F_2 above can be

- function
- arbitrary value of 'h'
- none of given choices

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Whe we apply Simpson's 3/8 rule, the number of intervals n must be

Answer (Please select your correct option)

- Even ✗
- Odd
- Multiple of 3 R ✓ Similarly in deriving composite Simpson's 3/8 rule, we divide the interval of integration into n sub-intervals, where n is divisible by 3, and applying the integration formula
- Multiple of 8

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 1/3 rule, valid number of intervals are.....

Answer (Please select your correct option)

7

8 ✗

5

3 R ✓

Thus Simpson's 1/3 rule is based on fitting three points with a quadratic.
Similarly, for n = 3, the integration is found to be

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

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Question No : 29 of 52

Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_0^1 e^{2x}dx$ by dividing into eight equal parts, width of the interval should be.....

Answer (Please select your correct option)

- 0.250
- 0.500
- 0.125
- 0.625

interval = 0 to 1

so

R ✓ $\frac{1-0}{8}$

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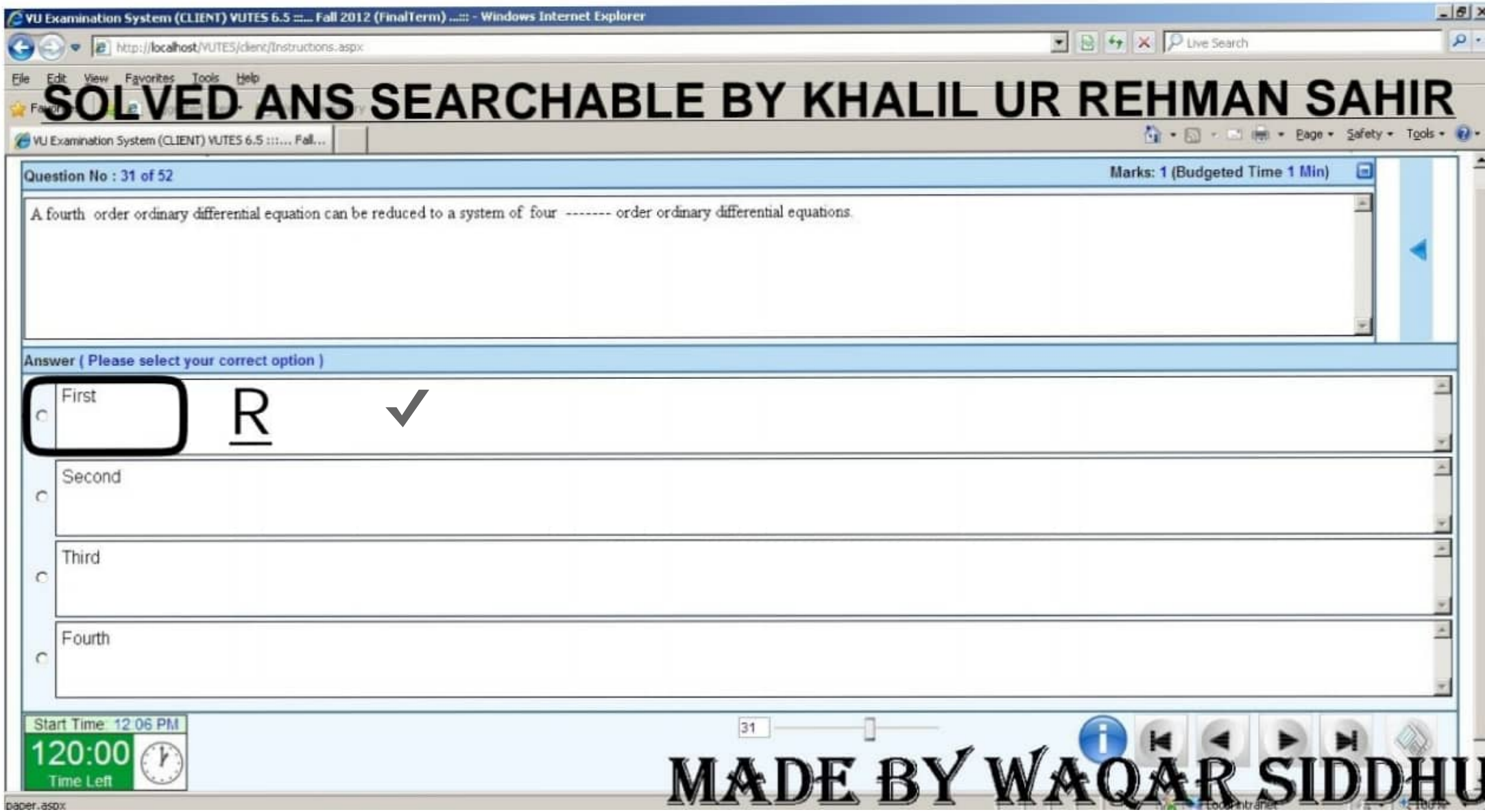

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$y' = x + y \; ; \; y(0.1) = 1.1$

$h = 0.1$, By Euler's method $y(0.2)$ is calculated as

Answer (Please select your correct option)

- 1.44
- 1.11
- 1.22 R ✓
- 1.33

$y' = x+y \; ; \; y(0.1) = 1.1 \; , h = 0.1$

$y(0.2) = ?$ Euler's Method $\rightarrow x_0 = 0.1$

$y_0 = 1.1$

$y_1 = y_0 + hf(x_0, y_0) = 1.1 + (0.1) f(0.1, 1.1)$

$y_1 = 1.1 + (0.1)(1.2) = 1.1 + 0.12$

$y_1 = 1.22$

$y(2) = 1.22$ Ans.

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

To solve the ordinary differential equation

$$3\frac{dy}{dx} + xy^2 = \sin x, y(0) = 5,$$

by Runge-Kutta 2nd order method, you need to rewrite the equation as

Answer (Please select your correct option)

- $\frac{dy}{dx} = \sin x - xy^2, y(0) = 5$
- $\frac{dy}{dx} = \frac{1}{3}(\sin x - xy^2), y(0) = 5$ R ✓
- $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{xy^3}{3}\right), y(0) = 5$
- $\frac{dy}{dx} = \frac{1}{3}\sin x, y(0) = 5$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_3 is given by

pg 205

Answer (Please select your correct option)

- $k_3 = hf(x_n + \frac{h}{3}, y_n + \frac{k_2}{3})$
- $k_3 = hf(x_n - \frac{h}{2}, y_n - \frac{k_2}{2})$
- $k_3 = hf(x_n - \frac{h}{3}, y_n - \frac{k_2}{3})$
- $k_3 = hf(x_n + \frac{h}{2}, y_n + \frac{k_2}{2})$ R ✓

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$y' = x^2 + 2xy \; ; \; y(0) = 1$

$h = 0.1$, By Euler's method $y(0.1)$ is calculated as

Answer (Please select your correct option)

1 R ✓

2

3

4

Q39 $y' = x^2 + 2xy \; ; \; y(0) = 1, \; h = 0.1 \longrightarrow y(0.1) = ?$ Euler's Method

$y_1 = y_0 + hf(x_0, y_0) = 1 + (0.1)(0) = 1$

$y_1 = \boxed{y(0.1) = 1}$

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_0^{\frac{\pi}{2}} \sin x dx$, by dividing the interval into four equal parts, width of the interval should be

xn-x0/n=pi/2-0/4=pi/8

Answer (Please select your correct option)

- π
- $\frac{\pi}{2}$
- $\frac{\pi}{4}$
- $\frac{\pi}{8}$ R ✓

We shall at first divide the range of integration $(0, \pi)$ into six equal parts so that each part is of width $\pi/6$ and write down the table of values:

X	0	π/6	π/3	π/2	2π/3	5π/6	π
Y=sinx	0.0	0.5	0.8660	1.0	0.8660	0.5	0.0

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

The minimum interval in which the root of the equation $x^2 - 3x + 1 = 0$ lie is

Answer (Please select your correct option)

- (0, 1)
- (1, 2)
- (2, 4)
- (0, 2)

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is .....

Answer (Please select your correct option)

- $\frac{(a+b)}{2}$ **R** **MID**
- $\frac{(a-b)}{2}$
- $\frac{(b-a)}{2}$
- None of the given choices

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Next approximation to the root of the given equation by bisection method can be found if

NO QUIESTION IS PROVIDED

Answer (Please select your correct option)

- $f(2) = -7,\ f(3) = -8$
- $f(2) = -7,\ f(3) = 8$
- $f(2) = 7,\ f(3) = 8$
- None of the given choices

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

If $8+7i$ is the root of a quadratic equation $f(x)=0$, then the other root of the given equation will be

Answer (Please select your correct option)

- $8-7i$ R ✓ MID
- $-8-7i$
- $-8+7i$
- 8

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

For the equation $x^3 + 3x - 1 = 0$, the root of the equation lies in the interval.....

MID

Answer (Please select your correct option)

- (1, 2)
- (1, 3)
- (0, 1)

R

- (1, 2)

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

The quantity of error which is present in the statement of the problem itself ,before finding its solution is called

Inherent errors
It is that quantity of error which is present in the statement of the problem itself, before finding its solution. It arises due to the simplified assumptions made in the mathematical modeling of a problem. It can also arise when the data is obtained from certain physical measurements of the parameters of the problem.

Answer (Please select your correct option)

- Inherent error
- Local round off error
- Local Truncation error
- Typing error

R ✓ MID TERM

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are two roots of the equation and one root is $2 + 3i$, then the other root will be

Answer (Please select your correct option)

- $2 + 3i$
- $2 - 3i$
- $3 + 2i$
- $3 - 2i$

R ✓ MID

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

It can be verified that for any matrix [A]

Answer (Please select your correct option)

- [A][A^{-1}]=[I] , I = identity matrix
- [A][A^{-1}]=[D] , D = diagonal matrix
- [A][A^{-1}]=[S] , S = symmetric matrix
- [A][A^{-1}]=[Z] , Z = orthogonal matrix

R ✓ MID

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In Crout's reduction method the elements of the main diagonal of upper triangular matrix are taken as

Crout's ReductionMethod

Here the coefficient matrix [A] of the system of equations is decomposed into the product of two matrices [L] and [U], where [L] is a lower-triangular matrix and [U] is an upper-triangular matrix with 1's on its main diagonal.

Answer (Please select your correct option)

- 0
- 1 R ✓ MID
- -1
- None of the given choices

None of the given choices

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

In Gauss-Seidal method, the new approximation calculated is

Note
The difference between jacobi's method and gauss Seidel method is that in jacobi's method the approximation calculated are used in the next iteration for next approximation but in Gauss-seidel method the new approximation calculated is instantly replaced by the previous one.

Answer (Please select your correct option)

- Used in the next iteration for next approximation
- Instantly replaced by the previous one ✓
- Never used again
- None of the given choices

R MID

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

In relaxation method, for fast convergence all the terms should be taken to one side and then reordering should be done so that the largest coefficients should appear on the

Solution
At first, we transfer all the terms to the right-hand side and reorder the equations, so that the largest coefficients in the equations appear on the diagonal.

Answer (Please select your correct option)

- End of the rows
- Beginning of the rows
- Diagonal
- None of the given choices

R ✓ MID

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Every square ------- matrix has an inverse.

Every square non-singular matrix will have an inverse.

Answer (Please select your correct option)

- Singular
- Non-singular
- Zero
- None of the above

R ✓ MID

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

If $S=\begin{pmatrix}1 & 2 & 3\\ 6 & 4 & 7\\ 8 & 6 & 5\end{pmatrix}$, then

Answer (Please select your correct option)

- $S^{-1} = S^{T}$ mid
- $S^{T} = S$
- $I = S$
- None of given choices R

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

A 3×3 matrix $[A]$ is said to be orthogonal if

Answer (Please select your correct option)

- $[A]^{-1}[A]=[I]$
- $[A]^{T}[A]=[I]$
- $[A]^{T}[A]=[O]$
- None of the given choices

mid

R

Definition

An $n \times n$ matrix $[A]$ is said to be orthogonal if

$[A]^{T}[A]=[I]$,

i.e. $[A]^{T}=[A]^{-1}$

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Jacobi's method for finding the eigen values can be applied to

In order to compute all the eigen values and the corresponding eigenvectors of a real symmetric matrix,

Answer (Please select your correct option) Jacobi's method is highly recommended.

- Real and symmetric matrix R ✓ mid
- Real and unsymmetric matrix
- Real and complex unsymmetric matrix.
- None of given choices.

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Lagrange's interpolation formula is used when the values of the independent variable are

Answer (Please select your correct option)

- Equally spaced
- Not equally spaced
- Constant
- None of the above

R

LAGRANGE'S INTERPOLATION FORMULA

Newton's interpolation formulae developed earlier can be used only when the values of the independent variable x are equally spaced. Also the differences of y must ultimately become small.

If the values of the independent variable are not given at equidistant intervals, then we have the basic formula associated with the name of Lagrange which will be derived now.

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are (n+1) values of y corresponding to (n+1) values of x, then we can represent the function f(x) by a polynomial of degree

Answer (Please select your correct option)

- n+2
- n+1
- n R ✓ , x_1, ...x_n . Since there are $(n + 1)$ values of y corresponding to $(n + 1)$ values of x, we can represent the function $f(x)$ by a polynomial of degree n.
- n-1

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	1	3	7
$f(x)$	2	4	10

Answer (Please select your correct option)

- Newton's forward difference interpolation formula
- Newton's backward difference interpolation formula
- Lagrange's interpolation formula R ✓
- None of the given choices

LAGRANGE'S INTERPOLATION FORMULA

Newton's interpolation formulae developed earlier can be used only when the values of the independent variable x are equally spaced. Also the differences of y must ultimately become small.

If the values of the independent variable are not given at equidistant intervals, then we have the basic formula associated with the name of Lagrange which will be derived now.

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	0	1	4
$y = f(x)$	2	1	4

$y(2)$

Answer (Please select your correct option)

- -1
- 1 ✓
- 2
- 4

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Relationship between differential operator and backward difference operator is given by

Answer (Please select your correct option)

- $hD = -1\log(1+\nabla)$
- $hD = 1\log(1+\nabla)$
- $hD = 1\log(1-\nabla)$
- $hD = -1\log(1-\nabla)$ R ✓

Using backward difference operator , we have $hD = -\log(1-\nabla)$.

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Shift operator is defined as

Answer (Please select your correct option)

- $Ef(x) = f(x-h)$
- $Ef(x) = -f(x-h)$
- $Ef(x) = -f(x+h)$
- $Ef(x) = f(x+h)$

R ✓

Shift operator, E

Let $y = f(x)$ be a function of x, and let x takes the consecutive values x, $x + h$, $x + 2h$, etc.
We then define an operator having the property

$$E f(x) = f(x+h)$$

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

using two-point equation the value of f '(0.5) is.....:

0.575-0.525/0.1=0.5

Answer (Please select your correct option)

- 0.73
- -0.75
- 0.5 R ✓
- -0.5

$$f'(0.1)=\frac{f(0.2)-f(0.1)}{h}=\frac{0.475-0.425}{0.1}=\frac{0.050}{0.1}=0.5.$$

$$f'(0.2)=\frac{f(0.3)-f(0.2)}{h}=\frac{0.400-0.475}{0.1}=\frac{-0.075}{0.1}=-0.75.$$

$$f'(0.3)=\frac{f(0.4)-f(0.3)}{h}=\frac{0.450-0.400}{0.1}=\frac{0.05}{0.1}=0.5.$$

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

$f''(0.2)$, using three-point equation will be calculated as .

MCQ INCOMPLETE

Answer (Please select your correct option)

17.5

12.5

7.5

-2.3

-12.5 **R**

$$f''(0.3) = \frac{f(0.4) - 2f(0.3) + f(0.2)}{h^2 = 0.01} = \frac{0.125}{0.01} = 12.5$$

$$f''(0.4) = \frac{f(0.5) - 2f(0.4) + f(0.3)}{h^2 = 0.01} = \frac{0.025}{0.01} = 2.5$$

$$f''(0.5) = \frac{f(0.6) - 2f(0.5) + f(0.4)}{0.01} = \frac{0.075}{0.01} = 7.5$$

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

The magnitude of truncation error in Milne's predictor formula is

Answer (Please select your correct option)

- $\frac{1}{90}h\Delta^4 y_0'$
- $\frac{1}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^3 y_0'$
- $\frac{28}{90}h\Delta^4 y_0'$

R ✓

From these equations, we observe that the magnitude of the truncation error in corrector formula is $1/90h\Delta^4 y_0'$ while the truncation error in predictor formula is $28/90h\Delta^4 y_0'$

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Let I denotes the closed interval spanned by $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, \overline{x}$. Then $F(x)$ vanishes ------times in the interval I.

Answer (Please select your correct option)

- 6
- 9 R ✓
- 7
- 8

Let I denotes the closed interval spanned by the values $x_0, \ldots, x_n, \overline{x}$. Then $F(x)$ vanishes $(n+2)$ times in the interval I.

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

From the following table of values:

x	1.3	1.5	1.7	1.9	2.1	2.3
y	2.9648	2.6599	2.3333	1.9922	1.6442	1.2969

Estimation of $y'(1.3)$ will be done using the formula of differential operator in terms of

Answer (Please select your correct option)

- Backward difference operator
- Central difference operator
- None of the given choices
- Forward difference operator

R ✓

Forward Differences

For a given table of values $(x_k, y_k), k = 0,1,2,...,n$ with equally spaced abscissas of a function $y = f(x)$, we define the forward difference operator Δ as follows

$\Delta y_i = y_{i+1} - y_i, \qquad i = 0,1,...,(n-1)$

To be explicit, we write

$$\Delta y_0 = y_1 - y_0$$
$$\Delta y_1 = y_2 - y_1$$
$$\vdots \quad \vdots \quad \vdots$$
$$\Delta y_{n-1} = y_n - y_{n-1}$$

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Trapezoidal rule, we approximate f(x) in the form .....

Answer (Please select your correct option)

- $ax+b$ R ✓
- ax^2+bx+c
- ax^3+bx^2+cx+d
- $ax^4+bx^3+cx^2+dx+e$

The Trapezoidal Rule (Composite Form)

The Newton-Cotes formula is based on approximating y = f (x) between (x_0, y_0) and (x_1, y_1) by a straight line, thus forming a trapezium, is called trapezoidal rule. In order to evaluate the definite integral

$$I = \int_a^b f(x)dx$$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Simpson's 3/8 rule, we approximate f(x) in the form .....

Answer (Please select your correct option)

- $ax+b$
- ax^2+bx+c
- ax^3+bx^2+cx+d R ✓
- $ax^4+bx^3+cx^2+dx+e$

Thus Simpson's 1/3 rule is based on fitting three points with a quadratic.
Similarly, for n = 3, the integration is found to be

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0+3y_1+3y_2+3y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Two segment trapezoidal rule of integration is exact for integrating at most ..... order polynomial.

Answer (Please select your correct option)

- first
- second
- third
- fourth

R

The Trapezoidal Rule (Composite Form)

The Newton-Cotes formula is based on approximating $y = f(x)$ between (x_0, y_0) and (x_1, y_1) by a straight line, thus forming a trapezium, is called trapezoidal rule. In order to evaluate the definite integral

$$I = \int_a^b f(x)dx$$

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 3/8 rule, the number of intervals can be....

Answer (Please select your correct option)

4 R ✓

Thus Simpson's 1/3 rule is based on fitting three points with a quadratic.
Similarly, for n = 3, the integration is found to be

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Romberg's integration method is ---------- than Trapezoidal and Simpson's rule.

Answer (Please select your correct option)

- none of the given choices
- more accurate
- less accurate
- equally accurate

R

ROMBERG'S INTEGRATION

We have observed that the trapezoidal rule of integration of a definite integral is of $O(h^2)$, while that of Simpson's 1/3 and 3/8 rules are of fourth-order accurate.

We can improve the accuracy of trapezoidal and Simpson's rules using Richardson's extrapolation procedure which is also called Romberg's integration method.

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The global error in Simpson's 3/8 rule is of

Answer (Please select your correct option)

- $o(h^2)$
- $o(h^3)$
- $o(h^4)$
- None of these choices

in Simpson's 1/3 rule, the global error is of $O(h^4)$.

We obtain the composite form of Simpson's 3/8 rule as

$$\int_a^b f(x)dx = \frac{3}{8}h[y(a) + 3y_1 + 3y_2 + 2y_3 + 3y_4 + 3y_5 + 2y_6 + \cdots + 2y_{n-3} + 3y_{n-2} + 3y_{n-1} + y(b)]$$

With the global error E given by

$$E = -\frac{x_n - x_0}{80}h^4 y^{(iv)}(\xi)$$

It may be noted that the global error in Simpson's 1/3 and 3/8 rules are of the same order.

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Rate of change of any quantity with respect to another can be modeled by

Answer (Please select your correct option)

- An ordinary differential equation R ✓

  Theories concerning electrical networks, bending of beams, stability of aircraft, etc., are modeled by differential equations.
  To be more precise, the rate of change of any quantity with respect to another can be modeled by an ODE

- A partial differential equation
- A polynomial equation
- None of the given choices

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

A third order ordinary differential equation can be reduced to a system of ---------- first order ordinary differential equations.

Answer (Please select your correct option)

0

1

2

3

R

For example, consider a second order differential equation of the form

$$y'' = f(t, y, y')$$

Introducing the substitution $p = y'$, the above equation reduces to a system of two first order differential equations, such as

$$y' = p, \qquad p' = f(t, y, p)$$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.02 at t=0.02. Using Euler's method, y at t= 0.04, h=0.02 is

h = t1 - t0

Answer (Please select your correct option)

- 3.0392 $y_1 = y_0 + hf(t_0, y_0),$
- 2.0392 $y_2 = y_1 + hf(t_1, y_1) = 1.02 + 0.02\frac{1.02-0.02}{1.02+0.02} = 1.0392$
- 1.0392 R ✓
- 0.0392

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_1 is given by

Answer (Please select your correct option)

- $k_1 = hf(x_n, y_n)$ R ✓
- $k_1 = 2hf(x_n, y_n)$
- $k_1 = 3hf(x_n, y_n)$
- None of the given choices

The fourth-order R-K method was described as

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_4 is given by

Answer (Please select your correct option)

- $k_4 = hf(x_n + 2h, y_n + 2k_3)$
- $k_4 = hf(x_n - h, y_n - k_3)$
- $k_4 = hf(x_n + h, y_n + k_3)$ R ✓
- None of the given choices

The fourth-order R-K method was described as

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The truncation error in Adam's predictor formula is

Answer (Please select your correct option)

- $\frac{251}{720}h\nabla^4 y_n'$ **R** This is known as Adam's predictor formula. The truncation error is $(251/720)h\nabla^4 y_n'$. ✓
- $\frac{251}{720}h\nabla^4 y_{n+1}'$
- $\frac{19}{720}h\nabla^4 y_{n+1}'$
- $\frac{19}{720}h\nabla^4 y_n'$

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x^2 + 2y \; ; \; y(1) = 3$$

$h = 1$, By Euler's method $y(2)$ is calculated as

Solution:

Given $y' = 2y + x^2, y(1) = 3, h = 1, y(2) = ?$

Answer (Please select your correct option)

Euler method

$$y_1 = y_0 + hf(x_0, y_0) = 3 + (1)f(1, 3) = 3 + (1) \cdot (7) = 3 + (7) = 10$$

$$\therefore y(2) = 10$$

wrong solution

- 4
- 6
- 8
- 10 R

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Obtain numerically the solution of

$$y' = x + y^2, \quad y(0) = 2$$

Using Euler's method to find y at $x = 1$, h=1

Answer (Please click here to Add Answer)

Solution:

Given $y' = x + y^2, y(0) = 2, h = 1, y(1) = ?$

Euler method

$$y_1 = y_0 + hf(x_0, y_0) = 2 + (1)f(0, 2) = 2 + (1) \cdot (4) = 2 + (4) = 6$$

$$\therefore y(1) = 6$$

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

If $f(0) = 3$ and $f(1) = 9$, then find the next approximate value of the function using secant method

Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a backward difference formula of $D^2 f(x)$.

Answer (Please click here to Add Answer)

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$$\frac{d}{dx} y_n = Dy_n = y_n'$$

$$= \frac{1}{h}\left(\nabla y_n - \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4} + \cdots \right)$$

$$y_n'' = D^2 y_n$$

$$= \frac{1}{h^2}\left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \cdots \right)$$

$$D = \frac{1}{h}\left(\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots \right)$$

$$D^2 = \frac{1}{h^2}\left(\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots \right)^2$$

$$= \frac{1}{h^2}\left(\nabla^2 + \nabla^3 + \frac{11}{12}\nabla^4 + \frac{5}{6}\nabla^5 + \cdots \right)$$

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a formula for Simpson's 1/3 rule.

$$\int_{x_0}^{x_2} f(x)dx = x_0y_0 + x_1y_1 + x_2y_2 + \text{Error} = \frac{h}{3}(y_0 + 4y_1 + y_2) - \frac{h^5}{90}y^{(iv)}(\xi)$$

Answer (Please click here to Add Answer)

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$$\int_{x_0}^{x_{2N}} f(x)dx = \frac{h}{3}[y_0 + 4(y_1 + y_3 + \cdots + y_{2N-1}) + 2(y_2 + y_4 + \cdots + y_{2N-2}) + y_{2N}] + \text{Error term}$$

This formula is called composite Simpson's 1/3 rule. The error term E, which is also called global error, is given by

$$E = -\frac{N}{90}h^5y^{(iv)}(\xi) = -\frac{x_{2N} - x_0}{180}h^4y^{(iv)}(\xi)$$

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Evaluate the integral

$$\int_{3}^{5} (\log x + 2)\,dx$$

Using Simpson's 3/8 rule
Take h=1

minimum four points are required for 3/8 rule. here three points

Answer (Please click here to Add Answer)

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h=1

x= 3 4 5

logx + 2=

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

This is known as Simpson's 3/8 rule, which is based on fitting four points by a cubic.

3/8 rule is not applicable

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write the two steps of solving the linear equations using Gaussian Elimination method

Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Runge-Kutta Method of order four to find the values of
k_1 and k_2 for the initial value problem

$$y' = yx^2, y(0) = 1, h = 0.1$$

Answer (Please click here to Add Answer)

Solution:

Given $y' = yx^2, y(0) = 1, h = 0.1$

Forth order R-K method

$$k_1 = hf\left(x_0, y_0\right) = (0.1)f(0, 1) = (0.1) \cdot (0) = 0$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = (0.1)f(0.05, 1) = (0.1) \cdot (0.0025) = 0.0003$$

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Trapezoidal rule
Take h=1

Answer (Please click here to Add Answ

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Trapezoidal rule

$\int_0^3 (x^2+x)\,dx \quad h=1$

$a=0 \quad b=3$

x	0	1	2	3
y	0	2	5	10

$$\int_0^3 (x^2+x)\,dx = \frac{h}{2}\left[y_0+y_3+2(y_1+y_2)\right]$$

$$= \frac{1}{2}\left[0+10+2(2+5)\right]$$

$$= \frac{1}{2}(24) = 12$$

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Question No : 49 of 52 — Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x+1)dx$$

Using Simpson's 3/8 rule

$$\int_{x_0}^{x_3} f(x)dx = \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3) - \frac{3}{80}h^5 y^{(iv)}(\xi)$$

Answer (Please click here to Add Answer)

Using Simpson's $\frac{3}{8}$ Rule

x	0	1	2	3
y	1	2	3	4

$$\int y dx = \frac{3h}{8}\left[(y_0 + y_3) + 2() + 3(y_1 + y_2)\right]$$

$$\int y dx = \frac{3 \times 1}{8}[(1 + 4) + 2 \times () + 3 \times (2 + 3)]$$

$$\int y dx = \frac{3 \times 1}{8}[(1 + 4) + 2 \times (0) + 3 \times (5)] = 7.5$$

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Find the dominant eigenvalue and the corresponding eigenvector of the matrix

$$\begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix}$$

by Power Method with vector $V^{(0)}=(1,1,1)^T$ as the initial vector.

Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

From the following table of values, construct backward difference table.

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Answer (Please click here to Add Answer)

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Newton's backward differentiation table is

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
0.1	1.10517			
		0.11623		
0.2	1.2214		0.01223	
		0.12846		**0.00127**
0.3	1.34986		**0.0135**	
		0.14196		
0.4	**1.49182**			

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^4 (x^2 + 1)dx$$

Using Simpson's 1/3 rule
Take h=1

Answer (Please click here to Add Answer)

x1 = 0 and x2 = 4
Step value (h) = 1

Solution:

Equation is $f(x) = x^2 + 1$.

The value of table for x and y

x	0	1	2	3	4
y	1	2	5	10	17

$$\int ydx = \frac{1}{3}[(1 + 17) + 4 \times (2 + 10) + 2 \times (5)]$$

$$\int ydx = \frac{1}{3}[(1 + 17) + 4 \times (12) + 2 \times (5)]$$

$$\int ydx = 25.3333$$

Solution by Simpson's $\frac{1}{3}$ Rule is 25.3333

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate the integral

$$\int_3^5 (\log x + 1)\,dx$$

Using Trapezoidal rule
Take h=1

Q42 $\int_3^5 (\log x + 1)\,dx$; $h = 1$ Trapzoidal rule

x	3	4	5
$y = f(x)$	1.4771	1.6020	1.6990

$$\Rightarrow \int_3^5 (\log x + 1)\,dx = \frac{h}{2}\left(y_3 + y_5 + 2y_4\right)$$

$$= \frac{1}{2}\left(1.4771 + 1.6990 + 2(1.6020)\right) = \frac{1}{2}(6.3801) = \boxed{3.19005} \text{ Ans}$$

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a formula for finding the value of k_1 in Fourth-order R-K method.

Answer (Please click here to Add Answer)

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$$k_1 = hf(t_n, y_n)$$

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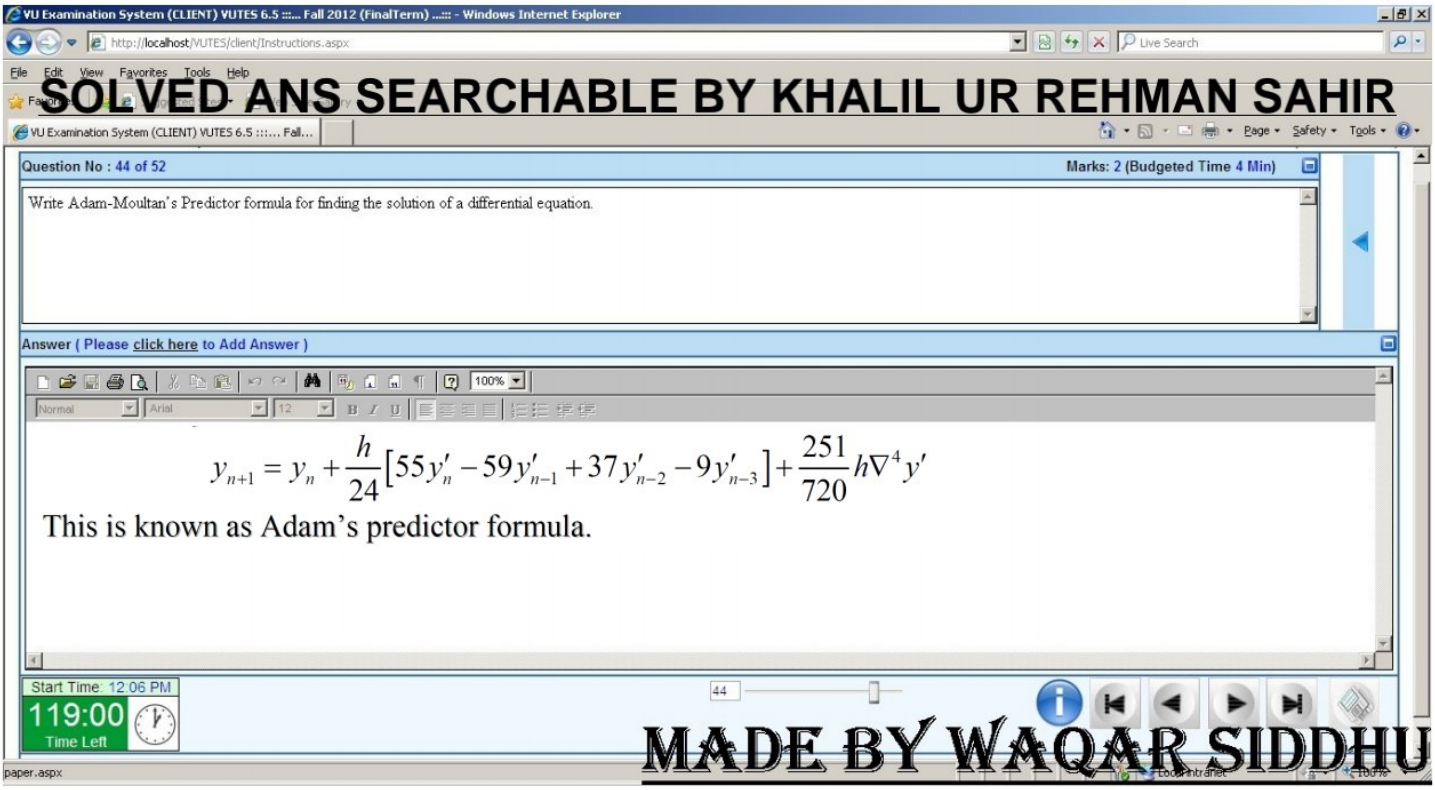

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Obtain numerically the solution of

$$y' = x^2 + 2x + y^2,\ y(0) = 1$$

Using Euler's method to find y at $x=1$, h=1

Answer (Please click here to Add Answer)

100%

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Q45 $y' = x^2 + 2x + y^2$; $y(0) = 1$ at $x = 1$, $h = 1$ Find y by Euler's Method

$y_1 = y_0 + hf(x_0, y_0) = 1 + (1)(1) = 2 \Rightarrow$ y_1 or $y(1) = 2$

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If, in solving a given differential equation, $y_0 = 1, y_1' = 1.1, y_2' = 1.2, y_3' = 1.3, h = 1$

Then find y_4 by Milne's Predictor formula.

$$y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3') + \frac{28}{90}h\Delta^4 y_0'$$

This is known as *Milne's predictor formula.*

Answer (Please click here to Add Answer)

Q47 $y_0 = 1, y_1' = 1.1, y_2' = 1.2, y_3' = 1.3, h = 1$ Find y_4 by Milne's predictor Formula.

$$y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3') = 1 + \frac{4(1)}{3}(2(1.1) - 1.2 + 2(1.3))$$

$$y_4 = 1 + \frac{4}{3}(3.6) = 1 + 4.8 = \boxed{5.8 = y_4}$$

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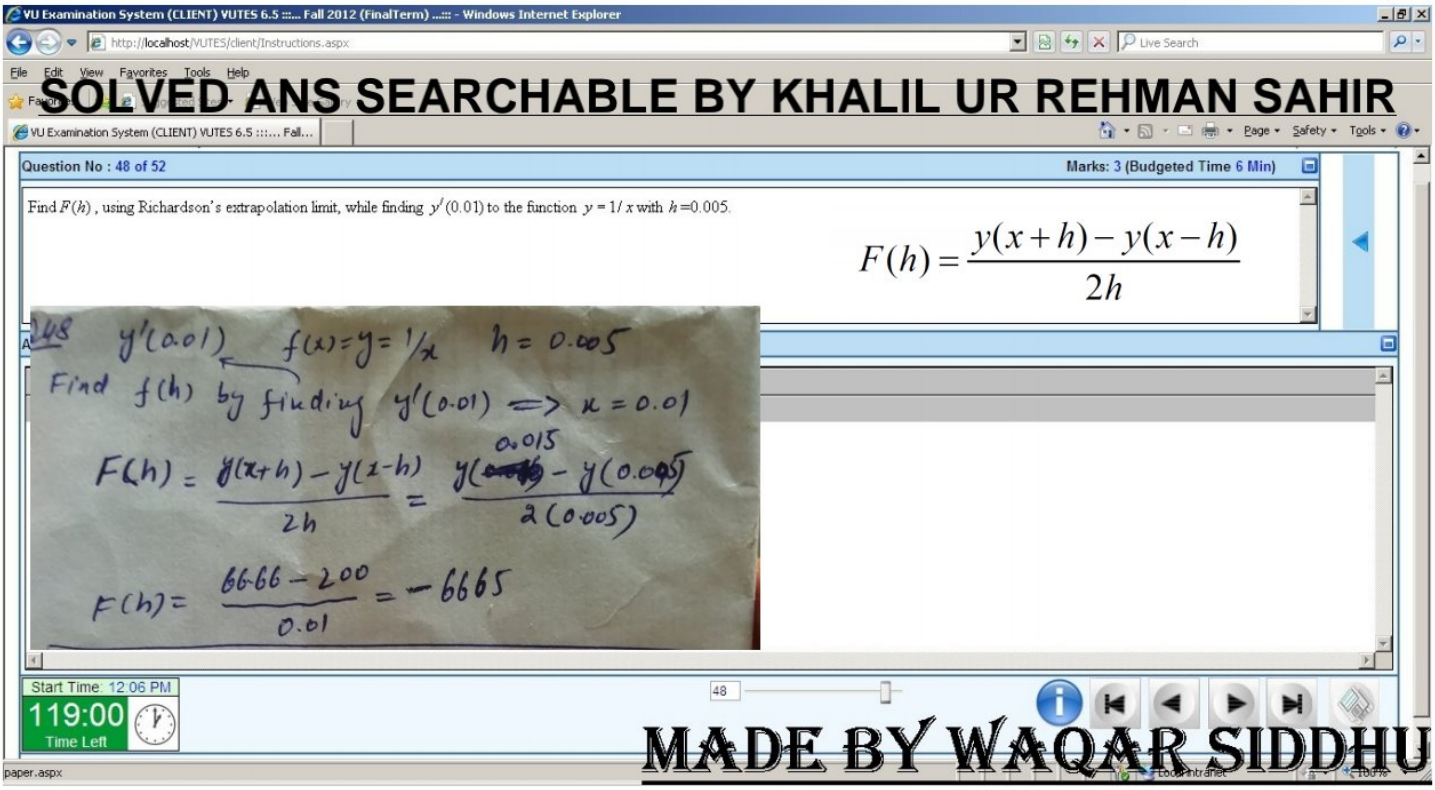

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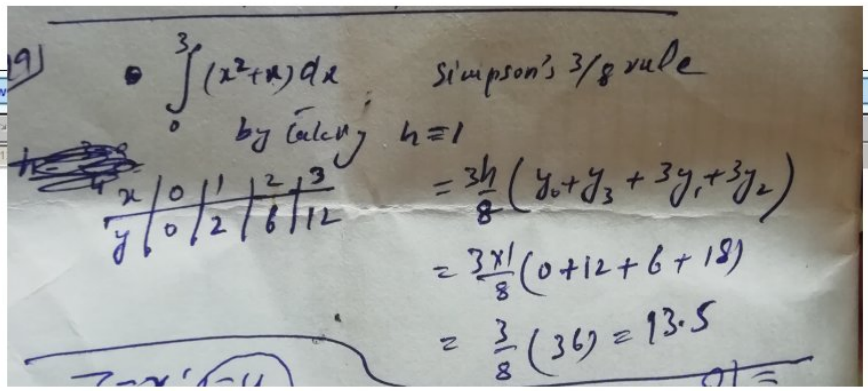
Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Simpson's 3/8 rule



Answer (Please click here to Add Answ

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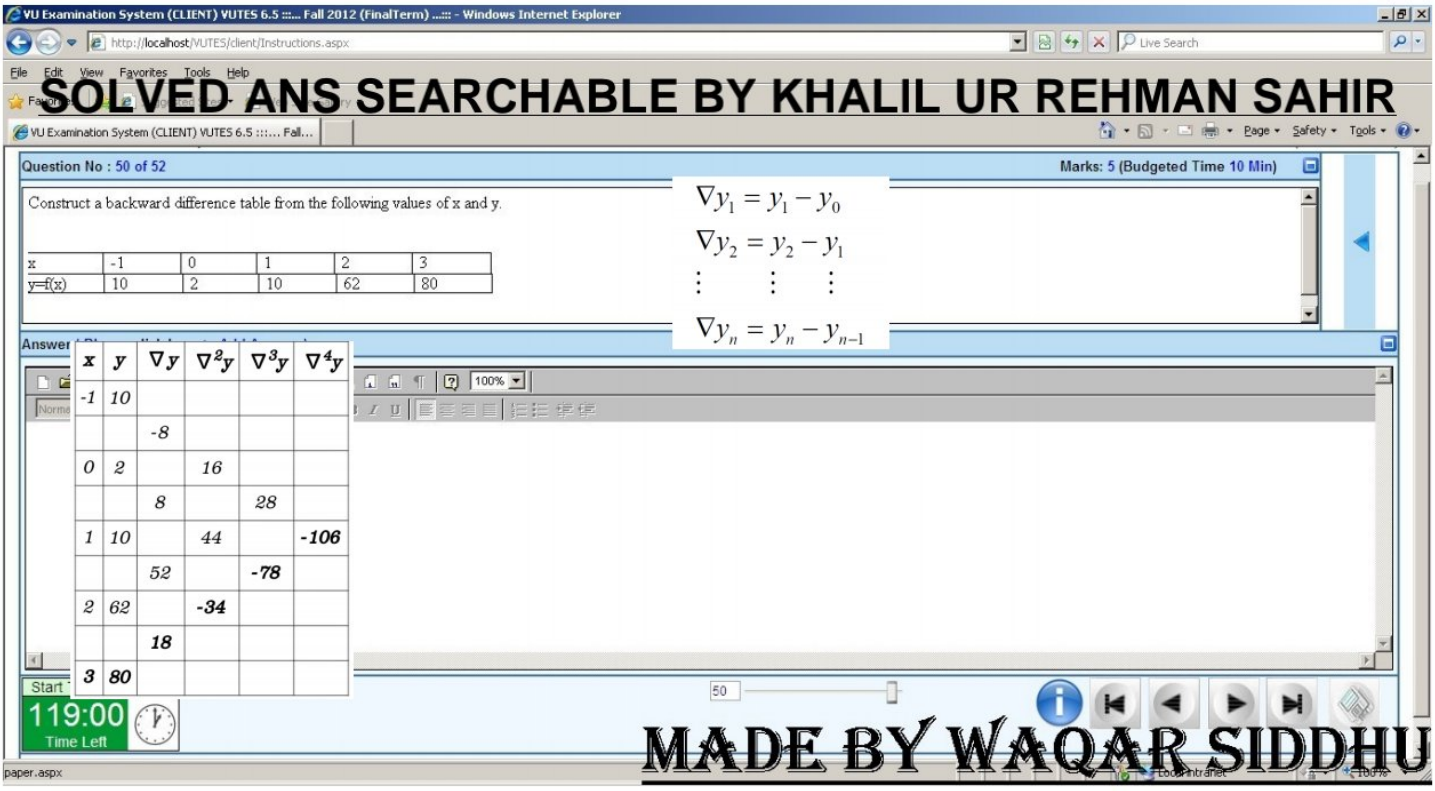

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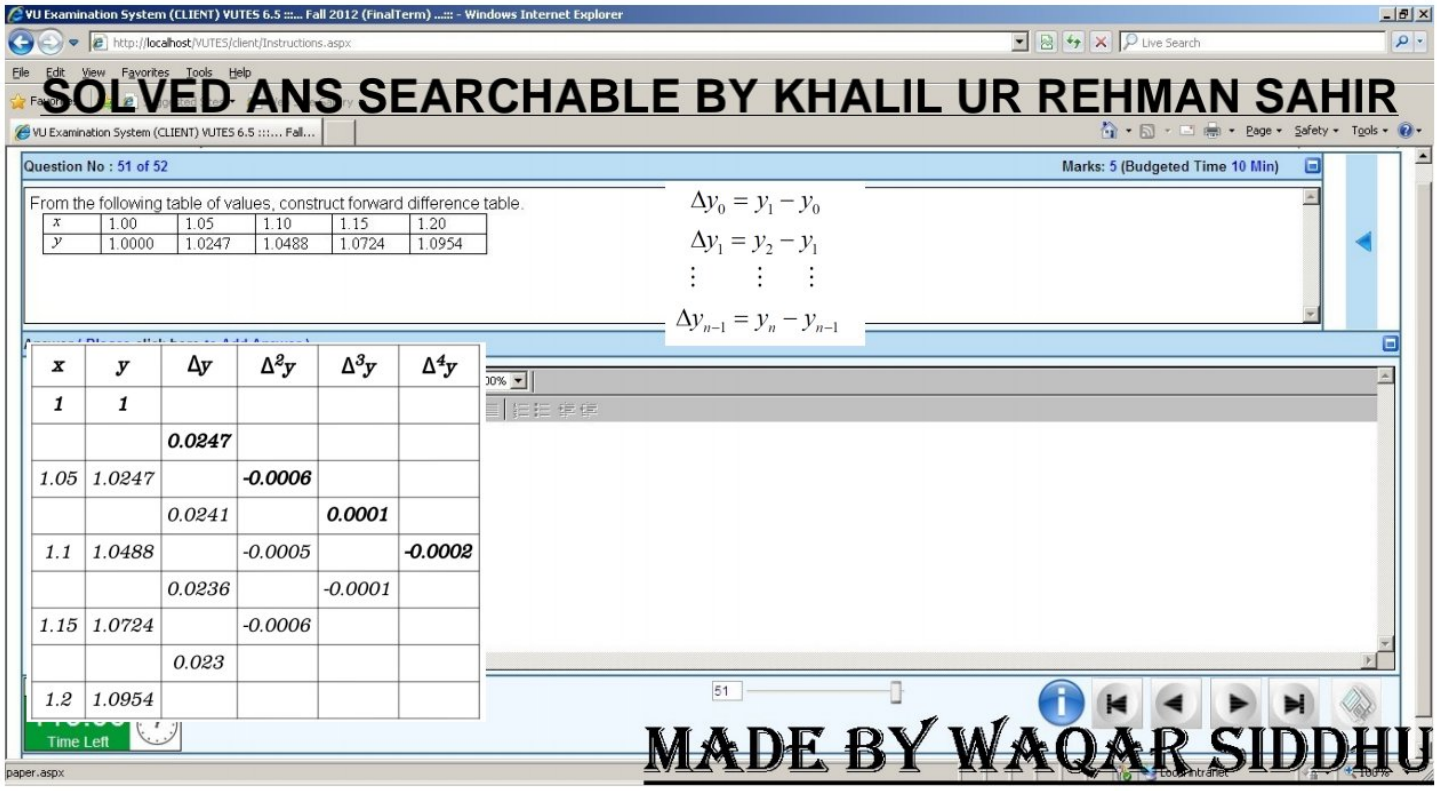

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	1				
		0.0247			
1.05	1.0247		-0.0006		
		0.0241		0.0001	
1.1	1.0488		-0.0005		-0.0002
		0.0236		-0.0001	
1.15	1.0724		-0.0006		
		0.023			
1.2	1.0954				

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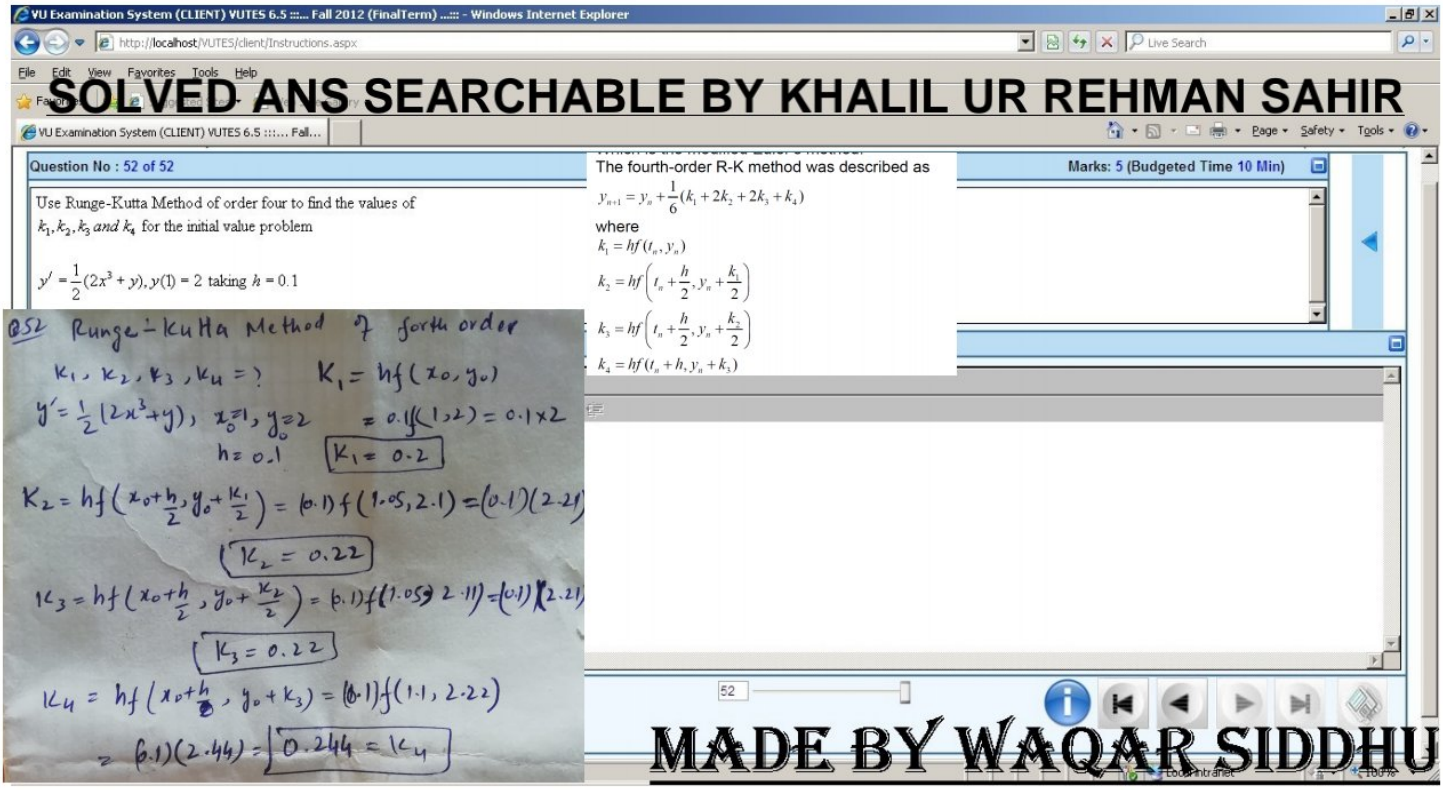

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

State the sufficient condition of convergence of the iterative solution to the exact solution

Answer

A sufficient condition for convergence of the iterative solution to the exact solution is

$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq 1}}^{n} |a_{ij}|, \quad i = 1, 2, ..., n$ When this condition (diagonal dominance) is true, Jacobi's method converges

A sufficient condition for convergence of iterative solution to exact solution is |a1| => (greater or equal) |b1|+|c1| |b2| => (greater or equal) |a2|+|c2| |c3| => (greater or equal) |a3|+|b3| For the system a1x+b1y+c1z=d1, a2x+b2y+c2z=d2, a3x+b3y+c3z=d3 Similarly for the system with more variables we can also construct the same condition

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

If $f(2) = -2.6146$ and $f(3) = 4.7610$, then find the first approximation using the Regula-Falsi method

Answer (Please click here to Add Answer)

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$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 3 - \frac{3-2}{4.7610 - (-2.6146)}(4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 3 - \frac{3-2}{4.7610 - (-2.6146)}(4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Simpson's 3/8 rule

Answer (Please click here to A

X	0	1	2	3
$Y=(x^2+x)$	0	2	6	12

$$\int_0^3 (x^2 + x)dx = \frac{3}{8}h\left[y_0 + y_3 + 3(y_1 + y_2)\right]$$

$$\int_0^3 (x^2 + x)dx = \frac{3(1)}{8}\left[0 + 12 + 3(2 + 6)\right]$$

$$\int_0^3 (x^2 + x)dx = \frac{3}{8}\left[12 + 24\right] = \frac{3}{8}(36) = \frac{108}{8} = 13.5$$

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MTH603

Numerical Analysis

File Version v11.02.02 *Prepared For: Final Term*

Note: Use Table Of Content to view the Topics, In PDF(Portable Document Format) format , you can check Bookmarks menu

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Compiled and Prepared by:

Muhammad Ishfaq (PakPattan)

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==========>

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Introduction To Numerical Analysis

Course Content:

Number systems, Errors in computation, Methods of solving non-linear equations, Solution of linear system of equations and matrix inversion, Eigen value problems, power method, Jacobi's method, Different techniques of interpolation, Numerical differentiation and integration, Numerical integration formulas, different methods of solving ordinary differential equations.

Each Topic One Solved Question with Steps:Compiled from Past assignment

Preliminary material:

Representation of numbers

Binary Number System (Base 2)=0,1

Used in Computer Internal Operations and Calculation

Octal Number System (Base 8)=0,1,2,3,4,5,6,7

ERROR

Types of error are,

- Inherent errors,
- Local round-off errors
- Local truncation errors

Decimal Number System (Base 10)=0,1,2,3,4,5,6,7,8,9

World Wide used and the most common system

Hexa Number System (Base 16)=0,1,2,3,4,5,6,78,9,A,B,C,D,E,F

Used in Processor Register Addressing

Errors in computations

The initial approximation

It may be found by two methods either by graphical method or
Analytical method
Graphical method

Non-Linear Equations

Bisection Method

Procedure in Detail:

Step,Take two Initial Approximation such that f(x1).F(x2)<0.Means both must have opposite signs.Take their mean by x3=(x1+x2)/2
Next Take two element from which 1 will be x3 and another from x1 or x2 such that both x3 and the other element should have opposite sign.
Repeat the above process to the required numbers of iterations.

Question 1

Find the root of the equation given below by bisection method.

$x^3 - x^2 + x - 7 = 0$

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(Note: accuracy up to three decimal places is required.) **Marks: 10**

SOLUTION

Let

$$f(x) = x^3 - x^2 + x - 7$$

$$\text{Now } f(1) = (1)^3 - (1)^2 + 1 - 7 = -6 < 0$$

$$f(2) = (2)^3 - (2)^2 + 2 - 7 = -1 < 0$$

$$\text{and } f(3) = (3)^3 - (3)^2 + 3 - 7 = 14 > 0$$

$$\text{As } f(2)f(3) = -14 < 0$$

Therefore a real root lies between 2 *and* 3

Iteration 1

Let

$x_0 = 2 \text{ and } x_1 = 3$

then

$$x_2 = \left(\frac{x_0 + x_1}{2}\right)$$

$$= \left(\frac{2+3}{2}\right) = 2.5$$

$$\text{Now } f(2.5) = (2.5)^3 - (2.5)^2 + 2.5 - 7 = 4.875 > 0$$

$$\text{As } f(2)f(2.5) = -1(4.875) = -4.875 < 0$$

Therefore a real root lies between 2 *and* 2.5

Iteration 2

$x_0 = 2 \text{ and } x_2 = 2.5$

then

$$x_3 = \left(\frac{x_0 + x_2}{2}\right)$$

$$= \left(\frac{2+2.5}{2}\right) = 2.25$$

$$\text{Now } f(2.25) = (2.25)^3 - (2.25)^2 + 2.25 - 7 = 1.578125 > 0$$

$$\text{As } f(2)f(2.25) = -1(1.578125) = -1.578125 < 0$$

Therefore a real root lies between 2 *and* 2.25

Iteration 3

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$x_0 = 2 \, and \, x_3 = 2.25$

then

$$x_4 = (\frac{x_0 + x_3}{2})$$

$$= (\frac{2 + 2.25}{2}) = 2.125$$

$Now \, f(2.125) = (2.125)^3 - (2.125)^2 + 2.125 - 7 = 0.205078 > 0$

$As \, f(2) f(2.125) = -1(0.205078) = -0.205078 < 0$

Therefore a real root lies between 2 *and* 2.125

Iteration 4

$x_0 = 2 \, and \, x_4 = 2.125$

then

$$x_5 = (\frac{x_0 + x_4}{2})$$

$$= (\frac{2 + 2.125}{2}) = 2.0625$$

$Now \, f(2.0625) = (2.0625)^3 - (2.0625)^2 + 2.0625 - 7 = -0.417725 < 0$

$As \, f(2.0625) f(2.125) = -0.417725(0.205078) = -0.085666 < 0$

Therefore a real root lies between 2.0625 *and* 2.125

Iteration 5

$x_5 = 2.0625 \, and \, x_4 = 2.125$

then

$$x_6 = (\frac{x_5 + x_4}{2})$$

$$= (\frac{2.0625 + 2.125}{2}) = 2.09375$$

$Now \, f(2.09375) = (2.09375)^3 - (2.09375)^2 + 2.09375 - 7 = -0.111481 < 0$

$As \, f(2.09375) f(2.125) = -0.111481(0.205078) = -0.022862 < 0$

Therefore a real root lies between 2.09375 *and* 2.125

Iteration 6

$x_6 = 2.09375 \, and \, x_4 = 2.125$

then

$$x_7 = (\frac{x_6 + x_4}{2})$$

$$= (\frac{2.09375 + 2.125}{2}) = 2.109375$$

$Now \, f(2.109375) = (2.109375)^3 - (2.109375)^2 + 2.109375 - 7 = 0.0455 > 0$

$As \, f(2.09375) f(2.109375) = -0.111481(0.0455) = -0.00507 < 0$

Therefore a real root lies between 2.09375 *and* 2.109375

Iteration 7

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$x_6 = 2.09375 \text{ and } x_7 = 2.109375$

then

$$x_8 = \left(\frac{x_6 + x_7}{2}\right)$$

$$= \left(\frac{2.09375 + 2.109375}{2}\right) = 2.1015625$$

$$\text{Now } f(2.1015625) = (2.1015625)^3 - (2.1015625)^2 + 2.1015625 - 7 = -0.0333 < 0$$

$$\text{As } f(2.1015625)\, f(2.109375) = -0.0333(0.0455) = -0.01515 < 0$$

Therefore a real root lies between 2.1015625 *and* 2.109375

Iteration 8

$x_8 = 2.1015625 \text{ and } x_7 = 2.109375$

then

$$x_9 = \left(\frac{x_8 + x_7}{2}\right)$$

$$= \left(\frac{2.1015625 + 2.109375}{2}\right) = 2.10546875$$

$$\text{Now } f(2.10546875) = (2.10546875)^3 - (2.10546875^2 + 2.10546875 - 7 = 0.006 > 0$$

$$\text{As } f(2.1015625)\, f(2.10546875) = -0.0333(0.006) = -0.0002 < 0$$

Therefore a real root lies between 2.1015625 *and* 2.10546875

Iteration 9

$x_8 = 2.1015625 \text{ and } x_9 = 2.10546875$

then

$$x_{10} = \left(\frac{x_8 + x_9}{2}\right)$$

$$= \left(\frac{2.1015625 + 2.10546875}{2}\right) = 2.103515625$$

$$\text{Now } f(2.103515625) = (2.103515625)^3 - (2.103515625)^2 + 2.103515625 - 7 = -0.01367 < 0$$

$$\text{As } f(2.103515625)\, f(2.10546875) = -0.01367(0.006) = -0.00008 < 0$$

Therefore a real root lies between 2.103515625 *and* 2.10546875

Iteration 10

$x_{10} = 2.103515625 \text{ and } x_9 = 2.10546875$

then

$$x_{11} = \left(\frac{x_{10} + x_9}{2}\right)$$

$$= \left(\frac{2.103515625 + 2.10546875}{2}\right) = 2.104492188$$

$$\text{Now } f(2.104492188) = (2.104492188)^3 - (2.104492188)^2 + 2.104492188 - 7 = -0.0038 < 0$$

$$\text{As } f(2.104492188)\, f(2.10546875) = -0.0038(0.006) = -0.0000228 < 0$$

Therefore a real root lies between 2.104492188 *and* 2.10546875

Iteration 11

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